Task 3 – Example Answer with Python Code

```
from sklearn.linear_model import LogisticRegression
from sklearn import metrics
import numpy as np
import pandas as pd

# Read in loan data from a CSV file
df = pd.read_csv('loan_data_created.csv')

# Define the variable features
features = ['credit_lines_outstanding', 'debt_to_income',
'payment_to_income', 'years_employed', 'fico_score']

# Calculate the payment_to_income ratio
df['payment_to_income'] = df['loan_amt_outstanding'] /
df['income']

# Calculate the debt_to_income ratio
df['debt_to_income'] = df['total_debt_outstanding'] /
df['income']

clf = LogisticRegression(random_state=0,
solver='liblinear', tol=1e-5,
max_iter=10000).fit(df[features], df['default'])
print(clf.coef_, clf.intercept_)

# Use the following code to check yourself
y_pred = clf.predict(df[features])

fpr, tpr, thresholds = metrics.roc_curve(df['default'],
y_pred)
print((1.0*(abs(df['default']-y_pred)).sum()) / len(df))
print(metrics.auc(fpr, tpr))
```